

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2024
Computer Networks (CS-317T)
Assignment # 2

Semester:V
Announced Date: 11-11-23
Total Marks: 3
Instructor Name: Razia Nisar Noorani

Batch: 2022F
Due Date: 29-11-24
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 2	Explain the services and functions provided by each layer in the Internet protocol stack	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)

Q#1: Write a Case study on Bit Torrent. Write detailed working with diagrams. Your Report may talk about the following questions.

1. How does the BitTorrent protocol work, and what makes it different from traditional file-sharing methods?
2. What role do "seeders" and "leechers" play in a BitTorrent network, and how do they contribute to the efficiency of file distribution?
3. Can you explain the concept of a "torrent file" in BitTorrent, and how it facilitates the sharing of large files across the network?
4. What are the potential advantages and disadvantages of using BitTorrent for file sharing compared to other methods like direct downloads or FTP?
5. How does BitTorrent ensure data integrity and reliability during the file-sharing process, especially in scenarios where files are split into multiple parts?
6. What measures does BitTorrent take to address issues related to copyright infringement, and how do content creators or distributors protect their intellectual property in a BitTorrent environment?
7. What are the common challenges or limitations associated with using BitTorrent, and how have developers and the community worked to overcome these issues?
8. How has the BitTorrent protocol evolved over the years, and what are some key improvements or changes that have been implemented to enhance its performance and security?
9. Can you discuss the role of decentralized tracking in BitTorrent and how it contributes to the scalability and resilience of the network?
10. What are some alternative uses or applications of BitTorrent technology beyond file sharing, and how has it been adopted in various industries?

Q#2: Write a report on Case Study of Netflix. Write detailed working with diagrams.

1. How has the rise of streaming services like Netflix transformed the traditional television and entertainment industry?
2. What are the key factors contributing to Netflix's success in the streaming market, and how does it differentiate itself from other platforms?
3. How does Netflix use algorithms and data analytics to personalize content recommendations for its users, and what impact does this have on user engagement?
4. In what ways has the production of original content, such as Netflix Originals, changed the landscape of television and film production globally?
5. How does Netflix handle regional content restrictions, and what challenges does the platform face in terms of licensing and distribution agreements across different countries?
6. Can you discuss the impact of binge-watching on viewer habits and the content consumption landscape, and how has Netflix influenced this trend?
7. What role does technology play in delivering high-quality streaming experiences on Netflix, and how has the platform adapted to advancements in internet infrastructure and devices?
8. How has Netflix addressed concerns related to data privacy and security, especially considering the vast amount of user data it collects for content recommendations?
9. What are the potential effects of the increasing competition in the streaming industry on Netflix, and how does the company continue to innovate to stay ahead?
10. How does Netflix approach the challenges of content discovery and retention, and what strategies does it employ to keep subscribers engaged over the long term?

NOTE:

1. **No** Assignment will be taken after due date.
2. Assignment can only be handed over to **CR**, If CR submitted assignment late it's his own responsibility no leniency will be given.
3. Only **Computerized** report will be submitted with proper file/ title page.
4. **4 members** can work together and submit one Report.